

Do *Polylepis australis* trees tolerate herbivory? Seasonal patterns of shoot growth and its consumption by livestock

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Abstract Browsing is one of the main factors determining survival, growth rate, woodland structure, and distribution of the high mountain tree *Polylepis australis*. This species has a substantial regrowth capacity, which may function as a mechanism to tolerate herbivory, but it is unknown to what extent it may compensate for the impact of herbivory. In 15 low-density tree stands subject to exclusion, moderate, and heavy livestock pressure, we selected 12 *P. australis* individuals <2 m tall, tagged four new shoots per tree and measured shoot length every month during a year. At the stand and at the tree level, we analyzed

monthly dynamics of growth and browsing, and the annual output in terms of total browsing and total gross and net growth (not discounting and discounting consumption, respectively). In addition, we assessed the influence of stand, tree and microsite characteristics on growth and browsing patterns. *Polylepis australis* fully compensated for herbivory in terms of shoot gross growth at moderate, but not at heavy livestock pressure. In terms of net growth, this species did not fully compensate for herbivory at any stocking rate. We found a strong coupling between browsing and growth along the year, suggesting that regrowth attracts browsing, and browsing promotes regrowth. At the stand level, annual gross growth was not affected by habitat characteristics, while at the tree level, annual gross growth decreased on more rocky microsites for browsed but not for unbrowsed trees. We concluded that stocking densities should be limited to allow for a reasonable annual net growth, as its nitrogen rich leaves are a valuable food resource and *P. australis* forests provide important ecosystem services.

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Introduction

Large herbivores can have substantial negative effects on tree species survival, growth rate, and growth form,

and consequently on forest structure and distribution (Hunter 1990). For example, in Scotland, birch woodland structure is strongly affected by large herbivores, and herbivore stocking rates highly influence woodland regeneration (Pollock et al. 2005). Browsing may negatively affect forest structure by producing mortality of seedlings and retarding the maturation of individuals (Danell et al. 2003; Motta 2003; Vandenberghe et al. 2007).

However, plants often have avoidance or tolerance mechanisms to cope with herbivory (Briske 1993; Rosenthal and Kothanen 1994; Katjiua and Ward 2006), especially in systems which have evolved under heavy herbivore pressure (Cingolani et al. 2005a). Woody plants often avoid consumption through chemical or physical defences which render them unpalatable (Massei et al. 2000; Gordon 2003). In these cases, herbivory can have an indirect positive effect on woody plants by eliminating competition or modifying fire regimes (Asner et al. 2004). In other cases, woody plants have developed tolerance to herbivory, achieved by compensating (at least partially) the lost tissues through a rapid regrowth (Belsky et al. 1993; Strauss and Agrawal 1999; Agrawal 2000; Stevens et al. 2007, 2008). In order to sustain rapid regrowth, plants need flexibility in allocation patterns, an architecture allowing lateral meristem activation, and tissues with high protein content, a trait which often increases palatability for herbivores (Chapin et al. 1993; Rosenthal and Kothanen 1994; Agrawal 2000; Stowe et al. 2000; Cingolani et al. 2005b). In some cases, plants even overcompensate the lost tissues (McNaughton 1979, 1983; Paige 1992; Lennartsson et al. 1998; Focardi and Tinelli 2005; Guillet and Bergström 2006), which has been interpreted as an expression of a mutualistic relationship between herbivores and plants (Agrawal 2000). While the benefit for herbivores of plant tolerance through compensation is clear, the benefit for plants from being eaten has been seriously questioned (Belsky et al. 1993; Stowe et al. 2000).

Overcompensating tolerance was defined as a higher biomass productivity (Belsky 1986) or a better fitness (Agrawal 2000; Stowe et al. 2000) of damaged (clipped or consumed) than undamaged plants, and has been considered as one extreme along a continuum of plant responses to consumer damage (Stowe et al. 2000). Following Belsky (1986), some authors compared the biomass produced by the damaged plants

(including the removed tissue which is part of the productivity) with biomass produced by the undamaged plants, considering productivity as a measure of fitness (e.g., Guillet and Bergström 2006). However, to compensate or over-compensate productivity does not necessarily mean that plants maintain or increase fitness (Noy-Meir 1993; Strauss and Agrawal 1999). In continuously grazed systems, where tolerant palatable plants are repeatedly consumed, overcompensating productivity can be associated with a loss of fitness because biomass regrowth is rapidly transferred to the next trophic level (Belsky 1986; Noy-Meir 1993), benefiting herbivores, but not plants. In woody species, net growth (i.e., the productivity which remains in the plant) can be a better measure of fitness than productivity (or gross growth) because seed production is highly dependent of plant size (Greene and Johnson 1994; Ordóñez et al. 2005). In this context, the simultaneous study of the effects of browsing on shoot gross growth, as a surrogate of aerial productivity and potential benefit for animals, and on shoot net growth, as a surrogate of plant fitness, could serve to obtain an integral view of the herbivore–plant interactions. This is particularly important when livestock is the main productive activity of an area, and woody plants are of conservation concern.

In the upper belt of the Córdoba Mountains (central Argentina), woodland cover has been seriously reduced due to four centuries of human activities (Renison et al. 2006; Cingolani et al. 2008). Local inhabitant's burn tussock grasslands and woodlands to produce a replacement by short grasses which are considered better forage for livestock (Pucheta et al. 1998; Renison et al. 2002). However, *Polylepis australis* Bitter, the dominant species of woodlands is highly consumed by livestock (Teich et al. 2005), although its fodder value is typically not acknowledged by animal owners (Pers. Commun.). Intensive browsing prevents height growth and retards woodland recovery after fires (Teich et al. 2005; Renison et al. 2006). Despite this, the species persists in many areas as small- or medium-sized shrubs, even under heavy livestock pressure, suggesting that it has tolerance mechanisms to compensate, and even over-compensate, for the lost tissues (Teich et al. 2005). Supporting this idea, an increased growth of this species after moderate fire has been observed (Renison et al. 2002).

Since *P. australis* woodlands are a priority for a conservation unit in these mountains (the National

Park Quebrada del Condorito and a surrounding Provincial Reserve on private lands), it is necessary to develop livestock management strategies (Administración de Parques Nacionales 2007). The management of the Reserve is a challenge because owners and conservation agencies have different objectives. However, both kinds of objectives can partially overlap if the potential of *P. australis* as a forage resource can be demonstrated to livestock owners. As a contribution to develop management strategies, our aim was to compare *P. australis* monthly and annual gross and net growth (as surrogates of biomass production and fitness, respectively), and browsing, at three levels of livestock pressure (exclusion, moderate, and heavy). Since the characteristics of the stand, and the characteristics of the tree and its surrounding biotic and abiotic microenvironment can strongly affect biomass production and browsing patterns (Bulinski and Mc Arthur 2003; Pietrzykowski et al. 2003; Heuze et al. 2005; Katjiua and Ward 2006; Lindroth et al. 2007), we included several of these factors into our analysis, which was performed at two levels: the stand and the tree level.

Methods

Study area

The study was carried out in the upper altitudinal belt (1,700–2,800 m a.s.l.) of the Córdoba mountains in Central Argentina (31° 34' S, 64° 50' W). Mean temperatures at 2,100 m a.s.l. of the coldest and warmest month are 5.0 and 11.4°C, respectively, with no frost-free period. Mean annual precipitation is 920 mm (1992–2000), with most rainfall concentrated in the warmer months, between October and April (Colladon 2000). The main economic activity is livestock rearing (mainly cattle, but also sheep, horses and goats), which began early in the seventeenth century. Additionally, some ethno-historical evidences suggest that before the introduction of European livestock the area was grazed by domestic camelids (*Lama glama*) raised by aboriginal people (Medina and Rivero 2009). Large native wild herbivores (*Lama guanicoe*, and probably *Rhea americana*) were locally extinct at the beginning of the twentieth century (Díaz et al. 1994).

Vegetation is a mosaic of tussock grasslands, grazing lawns, granite outcrops, evergreen *P. australis* woodlands, and eroded areas with exposed rock surfaces (Cingolani et al. 2004). Due to its intrinsic fragility and four centuries of domestic grazing and anthropogenic fires, the mountain range has now serious problems of erosion and woodland degradation (Cingolani et al. 2008). The present distribution of *P. australis* is mainly explained by the interaction between livestock-related impact and physical factors (Renison et al. 2006; Cingolani et al. 2008). At present, woodlands occupy 12% of the area, and are generally restricted to sites far from sources of human disturbances and on steep slopes in mid- to low-topographic positions, where fire and livestock have relatively low incidence (Cabido and Acosta 1985; Renison et al. 2002, 2004, 2006; Cingolani et al. 2004, 2008).

Sampling and data collection

We selected 15 *P. australis* stands (0.25–0.60 ha) with a low density of trees (<10% cover) of variable sizes and growth forms (trees and shrubs, Renison et al. 2006), distributed in the central and northern portion of the Córdoba mountain range (Teich et al. 2005). During the year of the study (from December 2004 to December 2005), six of those stands were under heavy livestock pressure, five under moderate livestock pressure, and four under livestock exclusion. The assignment of study stands to moderate or heavy livestock pressure treatments was performed by measuring dung frequency four times during the year of the study, and corroborated by direct observation of animals (as in Teich et al. 2005). In most cases, these stands had similar livestock pressure also during the previous two or more years (Teich et al. 2005; Teich 2006). One of the four stands without livestock was located in the National Park in a paddock where animals were excluded in the year 2000. The other three stands with no livestock were exclosures constructed in January 2004, located within three grazed paddocks (about 300 m² each stand).

In each stand, we selected 12 trees <2 m tall ($n = 180$) within the reach of livestock consumption. In January 2005, we tagged four representative very new shoots per tree, between 0.5 and 1.5 m above the ground level. Selected shoots had one or two

internodes, and the length of the shoots measured in January represent approximately the growth between December and January. From January on, we measured monthly the total length of the selected shoots, including lateral stems produced during the period. We also registered if the shoot was browsed or not. In many cases, tags disappeared, because animals ate the tag when browsing both the current and previous years more lignified biomass (i.e., biomass produced before December 2004, hereafter “old tissue”). In these cases, we registered the presence of old tissue consumption, and noted as zero the length of the selected shoots, since the true value (which should be negative), was unknown. In subsequent dates, the value remained at zero.

Additionally, at each stand, we measured altitude a.s.l.; an index of sun incidence (Renison et al. 2006) which mainly reflects topographic position; and the proportion of short grazing lawns, shrubby tussock grasslands, and rock (natural or exposed by erosion). These cover variables are related to the cumulative long-term livestock pressure of the stand (Cingolani et al. 2008), and describe quite well the present variability among stands (Teich et al. 2005). We also measured some variables at the tree level: height, shrubbiness (an index as in Teich et al. 2005), stem diameter, average browsing (% of stems browsed) of the previous 2 years (2003–2004; Teich et al. 2005; Teich 2006), an accessibility index (as in Teich et al. 2005), and rock (%), tussock grasses (%) and grazing lawn (%) in a 1 × 1 m area surrounding the tree.

Growth and browsing variables

We analysed a total of 174 trees because six trees lost all their tagged shoots at the beginning of the study due to uncertain causes. We calculated, for each originally tagged shoot, several monthly and annual growth and browsing variables. We used gross shoot growth (cm) as a surrogate of aerial biomass production and length of the shoot that was browsed (cm) as a surrogate of biomass consumption (Renison et al. 2002, A.M. Cingolani, unpubl. data). All variables were first calculated for each originally tagged shoot, then averaged per tree, and finally per stand. These averages were used for the tree and the stand level analyses, respectively. Thus, growth and browsing variables are expressed in cm per shoot determined between December 2004 and January

2005, or in percentage. Consequently, all measures are indicative of relative growth or browsing of current year tissues. For each shoot, we estimated the following variables:

Monthly gross growth (cm/shoot)

The length measured in January 2005 was assumed equivalent to the growth between December 2004 and January 2005, while for the other monthly periods, non-browsed shoot growth was estimated as the difference between the end and the beginning of the period. When browsing occurred, the shoot was generally shorter at the end than at the beginning of the period. In these cases, to calculate gross growth (and browsing, see below), we assumed that the shoot was browsed when it had produced half of the expected gross growth. The expected gross growth was estimated for each period independently as a proportion (calculated from all non-browsed shoots of the study) of the initial length. A preliminary data analysis of non-browsed shoots proved that this was a reasonable assumption. For the few cases of browsed shoots which were larger at the end than at the beginning of the period, gross growth was estimated as the difference between the end and the beginning, plus the browsed tissue which was assumed as a minimum value of 0.5 cm.

Monthly and annual cumulative gross growth (cm/shoot)

For each period, we calculated the cumulative gross growth by adding all the monthly growth values of the current and previous periods. For the last month (December), this measure was equivalent to the total annual gross growth.

Monthly and annual cumulative net growth (cm/shoot)

For each period, cumulative net growth was the shoot length remaining in the tree. When animals consumed old tissues, this value should be negative, but, as mentioned before, it was registered as zero. Therefore, in these cases, cumulative net growth was overestimated when averaged per tree and stand. For the last date (December 2005), this variable represented the annual net growth, and was equivalent to

the difference between annual gross growth and annual browsing.

Monthly browsing (cm/shoot)

For the period between December 2004 and January 2005 browsing was assumed as zero, because neither the shoots selected for the study, nor other similar very new shoots were found browsed. For the other periods, browsing was estimated as zero for non-browsed shoots, and as the difference between the beginning and the end of the period for browsed shoots, plus half of the expected growth for the period (see before). When the browsed shoot was larger at the end than at the beginning of the period, we assumed browsing as 0.5 cm. When animals consumed both old and current year tissue, our estimation represents only the current year tissue consumption.

Monthly and annual cumulative browsing (cm/shoot)

For each period, this variable was estimated as the sum of the current and previous periods' browsing. Thus, we assumed that the cumulative browsing at the end of our study (December 2005) represents the annual browsing of current year tissues.

Annual browsing intensity (%)

It is the proportion of the tissue produced during a year which was consumed during the same year. It was calculated as the ratio between annual browsing (cm) and annual gross growth (cm), expressed as a percentage.

Old tissue consumption (%)

For each shoot, we coded it as 1 if the tag had disappeared in any of the periods and as 0 if the tag had not disappeared. When averaged per tree or per stand and multiplied by 100, this variable represents the percentage of old tissues consumed during the year of the study.

Browsed shoots (%)

For each shoot, we coded it as 1 if it had been browsed anytime during the year and as 0 if not. As

before, this variable was multiplied by 100 after averaging per tree or stand.

Code of month of first browsing

For each shoot, we coded the date when it was browsed for the first time as follows: (1) browsed in the period between January and February, (2) between February and March, and so on, till 12, for shoots that in December were not yet browsed.

Stand level analyses

We compared the following annual growth and browsing variables between treatments (exclusion, moderate, and heavy livestock pressure): annual gross growth (cm/shoot), annual net growth (cm/shoot), annual browsing (cm/shoot), browsing intensity (%), old tissue consumption (%), browsed shoots (%), and code of month of first browsing. Additionally, we plotted for each treatment the cumulative gross growth (cm/shoot), the cumulative net growth (cm/shoot), and the cumulative browsing (cm/shoot) as a function of time, to visualize the seasonal patterns of growth (gross and net) and browsing. After a visual analysis of the seasonal curves, we calculated some additional variables to synthesize the growth and browsing seasonal dynamics and perform statistical comparisons between treatments (see Table 2 for details). All comparisons were done with General Linear Models, with the treatment as a fixed factor and the measured stand characteristics as continuous variables.

Livestock was excluded for a period of 3 months (March–June) from one of the stands (4) under heavy livestock pressure. Thus, we discarded stand 4 from the statistical analyses at the stand level.

Tree level analyses

In order to analyse growth and browsing patterns at the tree level, we performed two principal component analyses (PCA). First, we considered trees under livestock exclusion and used only monthly cumulative gross growth as variables. We excluded a few trees ($N = 6$) which were substantially browsed ($>20\%$, see results for details). In this way, we obtained the main patterns of variation in annual gross growth for non-browsed trees ($N = 40$). For

Table 1 Mean annual growth and browsing variables for grazing exclusion, moderate, and heavy livestock pressure treatments; and relationships with continuous variables

	Exclusion	Moderate	Heavy	Continuous variables	r^2
Annual gross growth (cm/shoot)	14.82 a	13.89 a	10.30 b	ns	0.63
Annual net growth (cm/shoot)	13.78 a	4.85 b	1.30 c	ns	0.97
Annual browsing (cm/shoot)	1.04 b	9.05 a	9.00 a	ns	0.84
Annual browsing intensity (%)	8.4 c	64.6 b	85.48 a	lawn (–)	0.97
Old tissue consumption (%)	6.3 c	44.6 b	68.9 a	lawn (–)	0.94
Browsed shoots (%)	14.2 b	85.4 a	97.9 a	ns	0.92
Code of month of the first browsing ^a	10.8 a	4.18 b	2.86 c	ns	0.91

Different letters indicate significant differences between treatments ($P < 0.05$)

ns not significant

^a Code of month of the first browsing: from 1 (browsed between January and February) to 12 (not yet browsed in December)

Table 2 Mean seasonal growth and browsing variables for livestock exclusion, moderate, and heavy livestock pressure treatments; and relationship with continuous variables

	Exclusion	Moderate	Heavy	Continuous variables	r^2
December–January gross growth (cm/shoot)	5.23 b	6.76 a	5.57 b	lawn (–) rock (–)	0.77
January–April gross growth (cm/shoot)	5.21 a	5.26 a	4.15 a	ns	ns
April–October gross growth (cm/shoot)	0.74 a	0.62 a	0.40 a	ns	ns
October–December gross growth (cm/shoot)	3.64 a	1.24 b	0.12 c	ns	0.91
Net growth till July (cm/shoot)	10.23 a	5.90 ab	2.98 b	ns	0.74
January–April browsing (cm/shoot)	0.50 b	5.20 a	4.91 a	ns	0.60
April–October browsing (cm/shoot) ^a	0.41 b	2.83 a	3.76 a	ns	0.57
October–December browsing (cm/shoot)	0.14 b	1.01 a	0.33 ab	ns	0.46

Different letters indicate significant differences between treatments ($P < 0.05$)

ns not significant

^a Square-root transformed for the analysis to meet statistical assumptions

trees under heavy and moderate livestock pressure treatments ($N = 128$), we performed a PCA using monthly cumulative browsing and gross growth variables together because, as seen in the stand level analysis, both patterns were coupled. In order to enhance the interpretation of PCA axes, we additionally correlated Axes 1 and 2 of both PCAs with the same synthetic variables used in the analyses at the stand level (Tables 1, 2).

The principal component analyses (PCA) Axes 1 and 2 of both analyses were new variables which summarized the seasonal patterns of growth and browsing. In order to depict which factors affect these patterns, we used General Linear Models, with PCA

Axes as dependent variables. As independent variables, we included “stand” as a random factor, and the measured tree and micro-site characteristics as continuous variables.

Results

Stand level analyses

Polylepis australis annual gross growth under livestock exclusion and moderate livestock pressure was around 14 cm/shoot, while under heavy livestock pressure it was significantly reduced to 10.3 cm/shoot.

Annual net growth was similar to annual gross growth under livestock exclusion, but it was strongly reduced even at moderate livestock pressure (Table 1). Annual browsing of current year tissue was only 1 cm/shoot under livestock exclusion, and rose to about 9 cm/shoot under moderate and heavy livestock pressure. Annual browsing intensity, in turn, differed among the three treatments, from 8% in livestock enclosures to 85% in sites under heavy livestock pressure. The same trend was found for the proportion of old tissue consumption, which varied from 6 to 69%; and for the proportion of current year shoot browsing, which varied from 14 to 98%; although, in this case, the differences were significant only for the exclusion treatment (Table 1). Under heavy livestock pressure, shoots started to be browsed at the beginning of the year (March or April), while under moderate livestock pressure, this happened 1 or 2 months later (May or June). The few browsed shoots observed in the exclusion treatment were located in the border of the enclosures at the reach of livestock, or in the paddock without grazing, where goats entered accidentally and stayed for a short period. Stand characteristics were not associated with annual growth and browsing variables in most cases. As an exception, the proportion of lawns showed a negative relationship with browsing intensity and old tissue consumption (Table 1).

We observed congruent seasonal patterns of cumulative gross and net growth in the exclusion treatment, reflecting their low browsing rates (Fig. 1a). In contrast, under moderate and heavy livestock pressure both curves diverged. While gross growth maintained a general increasing shape, cumulative net growth showed a peak and then a decrease coupled with the increasing cumulative browsing (Fig. 1b, c). The strong differences between treatments in the net growth curves were statistically corroborated not only for the annual output (Table 1) but also when comparing the net growth till July (Table 2). For gross growth, differences between treatments were less evident, but significant in some cases. At the beginning of the sampling period (December–January), gross growth was highest under moderate livestock, while at the end (October–December), gross growth was highest in the enclosures and very low under heavy livestock pressure (Table 2, Fig. 1). For the remaining periods of the year (January–April

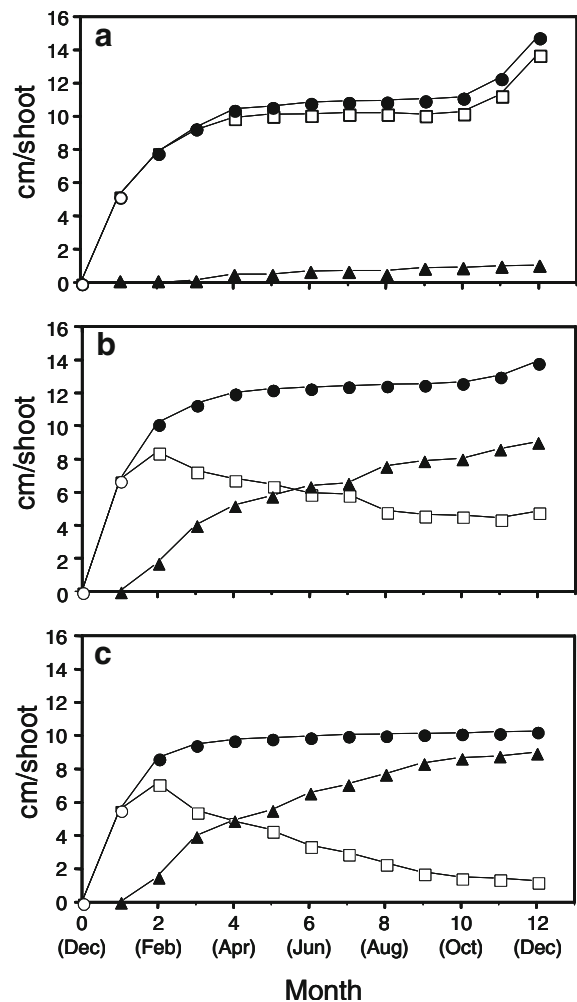


Fig. 1 Cumulative gross growth (filled circle) net growth (open square), and browsing (filled triangle) along the study period, for treatments with (a) grazing exclusion, (b) moderate, and (c) heavy livestock pressure

and April–October), gross growth was similar in all treatments. In contrast, treatments differed noticeably and significantly in the seasonal patterns of cumulative browsing (Fig. 1, Table 2). Until October, trees under exclusion experienced lower browsing (cm/shoot) than trees in both livestock treatments which were similar among them. At the start of the new growth season (October–December), browsing was highest in the moderate livestock pressure treatment and lowest in the enclosures (Table 2). Continuous stand variables in most cases were not significantly associated with descriptors of the coupled seasonal pattern of growth and browsing, except for

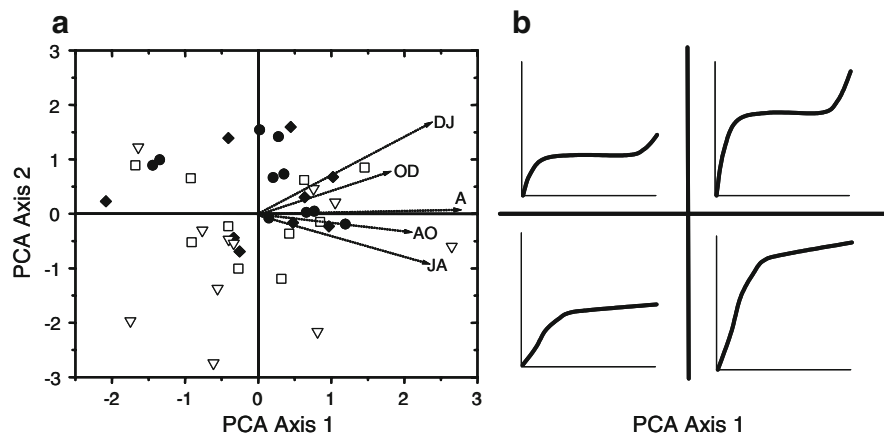


Fig. 2 PCA performed with cumulative monthly growth values for the 40 trees under livestock exclusion. **a** Different symbols indicate the different stands. Black symbols represent stands at higher altitudes above sea level (2,100–2,200 m a.s.l.) and open symbols represent stands at lower altitudes (1,800–

2,050 m a.s.l.). Arrows indicate correlation with annual gross growth (A) and gross growth at different periods: December–January (DJ), January–April (JA), April–October (AO), and October–December (OD). **b** For each quadrant of the plot, the average seasonal pattern is illustrated with a bold line

December–January growth, which decreased significantly with increasing proportion of grazing lawns and rock (Table 2).

Regarding the annual coupled curves of browsing and growth, and for most variables, stand 4 (with heavy livestock pressure but a release of 3 months), was more similar to the moderate than to the heavy livestock pressure treatment.

Tree level analyses

For the 40 trees under livestock exclusion, PCA Axes 1 and 2 explained 94 and 4% of total variance of cumulative monthly gross growth. Axis 1 reflected annual and monthly gross growth; trees with low growth had negative scores, while trees with high growth had positive scores (Fig. 2). Scores along Axis 1 were similar for the four stands under livestock exclusion, and were not related to any microsite or tree variable ($P \geq 0.05$). Axis 2 reflected the shape of the cumulative growth curve. Trees which began to grow early in the spring had positive scores, as indicated by the correlations of the axis with December–January and October–December gross growth. This was also visually corroborated by the steep slopes at the beginning and end of our measuring period (Fig. 2b). These earlier trees had lower growth from January to October when compared to trees at the negative side of axis 2. In summary, axis 2 separated earlier trees at the positive side, from later

trees at the negative side. Scores along axis 2 were significantly different between stands ($P \leq 0.05$). These differences suggested that trees at higher altitudes (black symbols in Fig. 2a) started to grow earlier than trees at lower altitudes. Additionally, trees that had been more browsed in the previous 2 years also began to grow early, as indicated by a significant positive relationship between this variable and the score along axis 2 ($P \leq 0.05$). For these trees under livestock exclusion during the year of the study, the average browsing of the two previous years varied from almost null to moderate (but never high) values.

For trees under moderate and heavy livestock pressure including stand 4, PCA Axes 1 and 2 using cumulative gross growth and browsing explained 75 and 14% of variance, respectively. Trees with high annual gross growth and browsing (cm/shoot) had positive scores along Axis 1, while trees with low total gross growth and browsing had negative scores (Fig. 3a). These annual patterns were defined mainly by the growth and browsing at the beginning of the study (from December to April for gross growth, and from January to April for browsing), as indicated by the correlations and the shape of the curves (Fig. 3b, c). Scores along Axis 1 were not different between stands ($P \geq 0.05$), but were positively related to tree height, and negatively with rock surrounding the individual ($P \leq 0.05$), indicating that taller trees growing in non-rocky microsites grew more and were more browsed. Axis 2 mainly reflects browsing

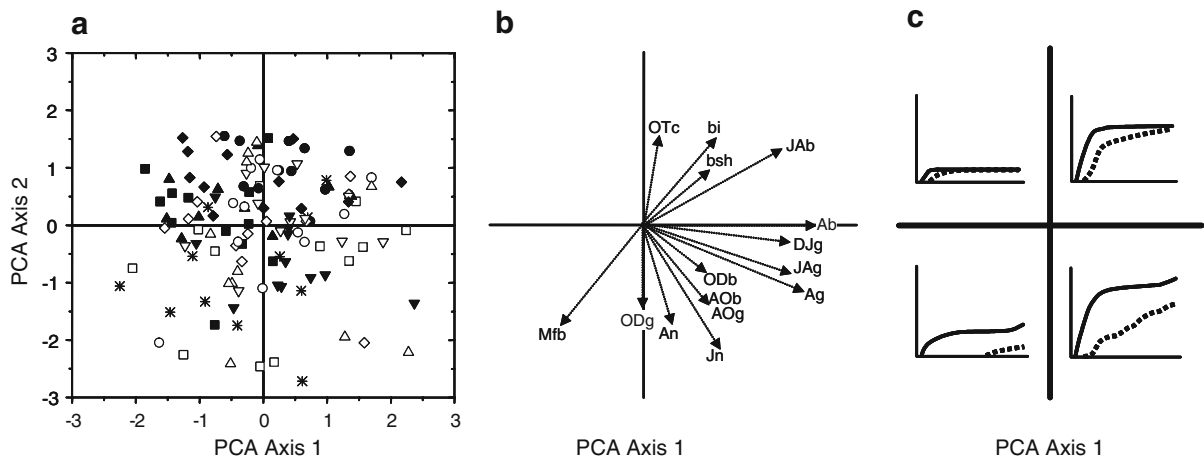


Fig. 3 PCA performed with cumulative monthly growth and browsing variables for the 128 trees under grazing. **a** Different symbols indicate the different stands. Black symbols represent stands under heavy livestock pressure, and open symbols represent stands under moderate livestock pressure. Asterisks represent stand 4. **b** Arrows indicate Axes' correlations with synthetic growth and browsing variables: annual gross growth (Ag), annual net growth (An), annual browsing (Ab), annual browsing intensity (bi), old tissue consumption (OTc), browsed

shoots (bsh), and month of the first browsing: (Mfb), December–January gross growth (DJg), January–April gross growth (JAg), April–October gross growth (AOg), October–December gross growth (ODg), net growth till July (Jn), January–April browsing (JAb), April–October browsing (AOb), and October–December browsing (ODb). **c** For each quadrant of the plot, the average seasonal pattern for growth is illustrated in solid lines, and the seasonal pattern of browsing in dotted lines

intensity. As shown by the correlations, heavily browsed (%) trees were located in the positive (upper) side, while less intensively browsed trees were located in the negative side. Additionally, trees in the negative side had higher net growth and slightly higher April–December and annual gross growth than trees in the positive side (Fig. 3b, c). Scores along Axis 2 differed between stands. Trees in stands with heavy livestock pressure tend to be in the positive side of the plot (black symbols in Fig. 3a), while trees in stands with moderate livestock are distributed along the whole range of Axis 2 scores (open symbols). Accessibility was also significant ($P \leq 0.05$), indicating that within a stand, more accessible trees were more heavily browsed.

Discussion

Does *P. australis* express tolerance to herbivory?

We found that *P. australis* is strongly consumed by domestic herbivores, a result in line with previous reports (Teich et al. 2005; Renison et al. 2006). At moderate livestock pressure, this species showed the

same shoot gross growth (as a surrogate of aerial productivity) as under livestock exclusion, expressing some degree of tolerance, or exact compensation, as defined by Belsky (1986). At heavy livestock pressure, repeated biomass removal impaired gross growth from current year tissues, precluding compensation of aerial productivity (“undercompensation”, according to Belsky 1986). However, the tolerance response in terms of shoot gross growth might be larger than reported due to resprouting from the base or from older stems, not only measured in our study but also observed in other species (Cierjacks and Hensen 2004; Persson et al. 2007). We have not measured root gross growth, but considering studies in other woody species (e.g., Weltzin et al. 1998; Stevens et al. 2008), and given the higher mortality of heavily browsed *P. australis* trees (Teich et al. 2005), it is probable that root growth was also impaired with heavy browsing. In contrast with gross growth, net growth was far lower for both livestock treatments than for livestock exclusion, because most current year growth was consumed. This indicates that under livestock pressure, *P. australis* trees are maintained small, as was also observed in previous studies (Teich et al. 2005). Since plant size is

associated with seed production in many species (Schmid and Weiner 1993; Greene and Johnson 1994), including *P. australis* (Pollice 2007), this result suggests that our study species does not compensate for herbivory in terms of fitness. Under heavy livestock pressure, the high consumption of old tissues suggests that true net growth is even lower than detected by our measures, and probably negative in several cases, as reported by Teich (2006). The strong consumption of *P. australis* despite its relatively low abundance in the stands contrasts with a much lower consumption of tussock grasses (Falczuk 2002) which are much more abundant in terms of coverage (Teich et al. 2005) but contain less nitrogen and probably high levels of lignin (Vendramini et al. 2000; Vandenberghe et al. 2007). This reinforces the idea that the evergreen *P. australis* is a selected item by livestock and a good food resource (Teich et al. 2005).

Seasonal patterns of browsing and growth

Initial growth (December–January) of *P. australis* was highest under moderate livestock pressure. Since this variable can be influenced by past livestock pressure (which was different than current year livestock pressure for 3 out of the 15 stands), we repeated this analysis considering previous year livestock pressure, and the result was reinforced (i.e., the differences were even stronger, results not shown). This is in line with the result obtained at the tree level for livestock exclosures, where trees moderately browsed in the previous 2 years started the growth period earlier than less browsed trees. These results suggest that previous browsing stimulates an earlier release of meristem winter dormancy. An architecture allowing for large number of meristems complemented with a rapid release of bud dormancy are known tolerance mechanisms (Stowe et al. 2000). Our results also suggest that browsing effects persist for some time after browsing release (Noy-Meir 1993), as shown by Guillet and Bergström (2006) for *Salix viminalis*.

The gross growth in the later periods of the year was similar for the three treatments, except at the beginning of the new growth season (October–December). Here, trees under heavy livestock pressure produced very little biomass from the current year shoots, since few of them were still present in October (36%, result not shown). This means that most buds produced during

any growth season are consumed before the start of the next. Part of the new season growth can start from older tissues or from the base, and thus the effect of heavy browsing can be less negative than evidenced by our results. However, resprouting from older tissues makes it more difficult to compensate for plant height, therefore, if heavy browsing persists, plants become progressively shorter, and eventually die, as shown by Teich et al. (2005). Nevertheless, we have observed very few of our study trees resprouting from the base, and this seems to happen only when a large proportion of the aerial biomass has been seriously damaged (M. A. Giorgis, personal observation) or burned (Renison et al. 2002).

Trees under heavy livestock pressure were browsed more intensively and earlier in the year than trees under moderate livestock pressure. However, in terms of tissue removed (cm), browsing was similar in both treatments, except at the beginning of the new growth season (October–December) when almost no current year shoot tissue was available (and hence browsed) under heavy livestock pressure. The strong coupling between gross growth and browsing patterns along the year was not detected in other studies, because they generally measure response to one or few browsing events without analyzing further browsing on regrowth tissues (e.g., Harmer 2001; Guillet and Bergström 2006; Bee et al. 2007). Our results suggest that regrowth tissues attract browsing, and browsing promotes regrowth if at least one bud remains active.

Influence of habitat and tree characteristics

Annual gross growth of *P. australis* was not affected by any stand variable, but December–January growth was affected negatively by the proportion of lawn and rock in the stand. This suggests that, in degraded and xeric habitats, the growth period starts later due to scarcity of soil moisture at the end of the dry season, when temperature begins to increase. In contrast to our results, Marcora et al. (2008) found a strong effect of environment on annual gross growth, probably because they analysed a pronounced altitudinal gradient. We also found that the proportion of lawns affects browsing intensity. Lawns are the most nutritive and selected plant community in the area (Pucheta et al. 1998; Cingolani et al. 2003), and given similar livestock pressures, the presence of this

community in the stand presumably alleviates the browsing pressure on *P. australis*.

At the tree level, neither did the annual gross growth under livestock exclusion (mainly reflected in PCA Axis 1, Fig. 2) differ between stands, nor was any microsite variable related to total gross growth. However, the timing of the growth period (reflected in Axis 2) under livestock exclusion was different depending on the stand. Trees in stands located at higher altitudes, showed an earlier start of the growth period. We attribute this to the higher soil water availability at high altitudes at the end of the dry period due to reduced evapotranspiration. As in the exclosures, under moderate and heavy livestock pressures stands did not differ in their trees' total gross growth (coupled with total browsing and reflected in PCA Axis 1, Fig. 3). However, in this case, we found that a small tree size and a rocky microsite negatively influenced gross growth. Total gross growth was also slightly reflected in PCA Axis 2, indicating that total gross growth was influenced by livestock pressure once the tree and microsite variability were accounted for, which is in line with our results at the stand level.

The microsite and tree characteristics influence *P. australis* gross growth under livestock pressure, but not under livestock exclusion is probably related to altered allocation of photoassimilates across plant organs in response to loss of photosynthetic tissues (Stevens et al. 2008). Under livestock pressure, taller trees produced more growth than smaller trees which might be the result of their higher storage capacity for energy resources and nutrients (McNaughton 1983; van der Heyden and Stock 1996). Trees of rocky microsites may have been less productive due to lack of soil nutrients and/or water. The importance of high availability of soil resources and plant reserves for compensating tolerance was highlighted by a number of authors (McNaughton 1983; Noy-Meir 1993; Belsky et al. 1993; Katjiua and Ward 2006; Stevens et al. 2007; Wise and Abrahamson 2007). Our results partially agree with those obtained by Renison et al. (2005) which found a decreased growth of *P. australis* planted seedlings in rocky microsites.

Management implications

Polylepis australis has a great regrowth capacity, non-effective physical or chemical defences and leaves relatively rich in nitrogen (Vendramini et al. 2000).

Probably due to these characteristics, it is highly selected and consumed by herbivores and could be an alternative food resource in the Argentine mountains, especially in winter when forage grasses become less available (Rao et al. 2003; Chakeredza et al. 2007; Pucheta et al. 1998). Our data imply that *P. australis* could be a valuable alternative forage resource under conditions of moderate livestock pressure, but only in areas where tall reproductive trees are still present. The challenge is how to balance the different land-use and conservation demands (Evans et al. 2006). Because *P. australis* consumption is so high even under moderate livestock pressure, it is important to manage livestock so as to let sufficient small trees escape browsing and reach reproductive stages. Thus, when elaborating management strategies, stocking densities should be limited to such extent that *P. australis* is able to tolerate herbivory and to produce some annual net growth. Areas where *P. australis* cover is very low should be excluded from livestock until woodlands have recovered. Further research on the association between tree shoot and root productivity, woody forage quality, and stocking rates is necessary to facilitate sustainable livestock management within mountain woodlands.

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